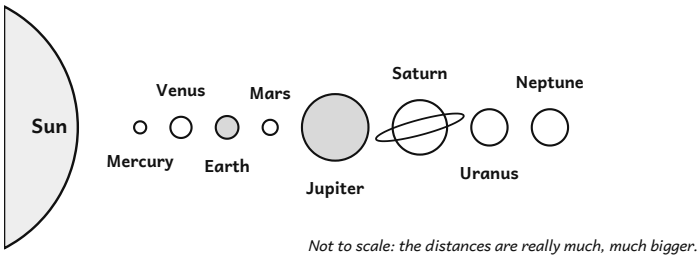
EARTH AND SPACE

# The Solar System

Stars, planets, moons, meteors and satellites, a model of the solar system, and scientists who studied the sky




*Not to scale: the distances are really much, much bigger.*

**In this pack**

- **Learn it:** Our solar system ★☆☆
- **Fact check:** Space objects ★☆☆
- **Think it through:** Thinking about space ★★☆
- **Investigate:** Build a model solar system ★★☆
- **Real-world case study:** Bangladesh in space: Bangabandhu Satellite-1 ★★★
- **Review and challenge:** Show what you know ★★★

**This pack belongs to:** _____ **Class:** _____

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LEARN IT · B1 · SOLAR SYSTEM

# Our solar system

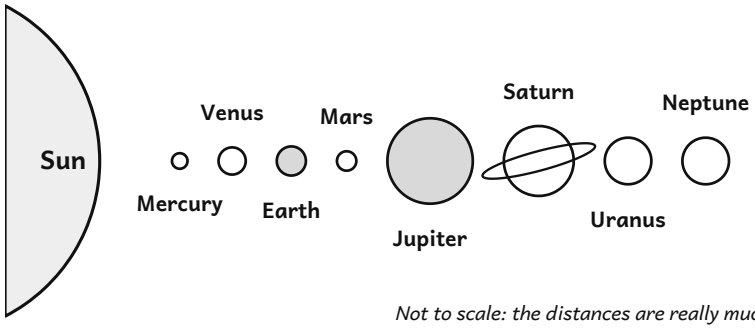
★☆☆ Grade 3

Name: Class: Date:

**Learning goals**

- Tell the difference between stars, planets, moons, meteors and satellites.
- Describe the solar system and make a model of it.
- Learn about scientists who studied the Sun, planets and stars.

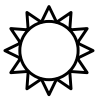
On a clear night, look up. You may see the Moon, thousands of stars and maybe a bright planet. The Earth is one of the planets that travel around our star, the **Sun**. Together they make the **solar system**.




*Not to scale: the distances are really much, much bigger.*

*The Sun and its eight planets (not to scale).*

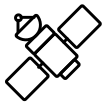
| Object | What it is | Example |
|---|---|---|
| **star** | a huge ball of hot, glowing gas that makes its own light | the Sun |
| **planet** | a large round object that travels around a star | Earth, Mars, Jupiter |
| **moon** | an object that travels around a planet | our Moon |
| **meteor** | a space rock that burns up in our air: a "shooting star" | meteors in August |
| **satellite** | anything that travels around a planet. People build **artificial** satellites for TV, phones and weather | Bangabandhu Satellite-1 |


star


planet


moon


meteor


satellite

## The eight planets

In order from the Sun: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune**. A way to remember: "**M**y **V**ery **E**asy **M**ethod **J**ust **S**peeds **U**p **N**aming." Jupiter

is the biggest. Saturn has beautiful rings. Earth is the only planet we know with living things.

### How it works

The planets travel around (**orbit**) the Sun. **Gravity** from the Sun keeps them in their paths. The Earth takes one **year** to go once around the Sun. The Moon orbits the Earth.

### Scientists who studied the sky

- **Aryabhata** (India, about 1500 years ago) said that the Earth spins, which makes the stars seem to move.
- **Copernicus** (Poland, 1543) said the planets go around the **Sun**, not around the Earth.
- **Galileo** (Italy, 1610) used a telescope and discovered four moons going around Jupiter.
- **Dr Jamal Nazrul Islam** (Bangladesh, 1939–2013) was a famous scientist from Chattogram who wrote about how the universe began and how it might end.

> ⚠ **Stay safe**
>
> Never look at the Sun, even through sunglasses or a telescope. It can damage your eyes.

> 🕮 **Key words**
>
| | |
|---|---|
| **solar system** | the Sun and everything that travels around it |
| **star** | a ball of hot glowing gas |
| **planet** | a large object orbiting a star |
| **moon** | an object orbiting a planet |
| **orbit** | the path around something |
| **satellite** | an object that orbits a planet |

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LEARN · PLAY · WONDER

FACT CHECK · B1 · SOLAR SYSTEM

# Space objects

★☆☆
Grade 3

**Name:** _____ **Class:** _____ **Date:** _____

**Learning goals**

- Name space objects.
- Order the planets.

## 1 Draw a line to match.

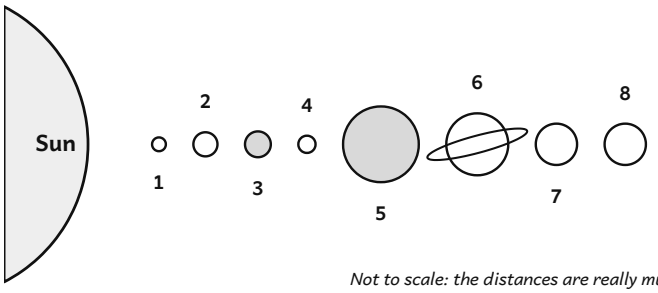
[5]

Draw a line to match each pair, or write the letter.

| | |
|---|---|
| star ● | ● **A** travels around a planet |
| planet ● | ● **B** a shooting star |
| moon ● | ● **C** sends TV and phone signals |
| meteor ● | ● **D** travels around a star |
| satellite ● | ● **E** a ball of hot glowing gas |

## 2 Write the planets in order from the Sun.

[8]




*Not to scale: the distances are really much, much bigger.*

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

## 3 True or false?

[3]

| Statement | True | False | If false, correct it |
|---|---|---|---|
| 1. The Sun is a star. | | | |
| 2. The Moon makes its own light. | | | |
| 3. Jupiter is the biggest planet. | | | |

THINK IT THROUGH · B1 · SOLAR SYSTEM

# Thinking about space

★★☆
Grade 3

Name: Class: Date:

**Learning goals**

- Explain how the solar system works.

**1** **Why does the Moon shine at night if it has no light of its own?** [2]

**2** **What keeps the planets going around the Sun?** [1]

- ○ wind
- ○ gravity
- ○ magnets

**3** **What did Copernicus say? Why was it a big new idea?** [2]

INVESTIGATE · B1 · SOLAR SYSTEM

# Build a model solar system

★★☆
Grade 3

**Name:** **Class:** **Date:**

**Learning goals**

- Make a model.
- Say what a model shows and what it cannot show.

**You will need**

balls of clay or paper in different sizes, string, card, glue, a big sheet of paper.

Make the Sun in the middle and put the eight planets in order around it. Make Jupiter and Saturn the biggest planets and Mercury the smallest.

**1** **Draw your model and label the planets.** [4]

My solar system model

**2** **What does your model show well?** [1]

**3** **What can't your model show?** [1]

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LEARN · PLAY · WONDER

# Bangladesh in space: Bangabandhu Satellite-1

★★★
Grade 3

**Name:** **Class:** **Date:**

**Learning goals**

- Learn how satellites help people.

Science | Technology | History

In **May 2018**, Bangladesh's first satellite, **Bangabandhu Satellite-1**, was launched into space on a rocket from the United States. It travels far above the Earth.

The satellite sends **TV** and **internet** signals to people all over Bangladesh, even on islands and in hills where it is hard to put cables. Weather satellites from other countries also help Bangladesh see cyclones coming.

Based on information about Bangabandhu Satellite-1.

**1** **What is an artificial satellite?** [1]

**2** **How does the satellite help people in Bangladesh?** [2]

**3** **Design your own satellite. What job would it do?** [2]

My satellite

REVIEW AND CHALLENGE · B1 · SOLAR SYSTEM

# Show what you know

★★★
Grade 3

Name: Class: Date:

**Learning goals**

- Use space words.

## 1 Find the words.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| G | E | V | M | M | V | V | J | J |
| E | A | E | L | N | O | T | U | W |
| P | R | M | H | R | M | O | C | H |
| L | T | V | S | U | N | E | N | W |
| A | H | E | M | K | J | O | T | U |
| N | L | W | H | S | T | A | R | P |
| E | J | O | R | B | I | T | U | R |
| T | W | C | K | G | T | J | I | O |
| S | M | A | R | S | K | K | T | U |

**earth mars moon orbit planet star sun**

## 2 How many planets are in our solar system? [1]

◯ 6 ◯ 8

◯ 10

## 3 A shooting star is really a... [1]

◯ star ◯ meteor

◯ planet

## 4 Challenge. Why do you think there is life on Earth but not on Mercury? [2]

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| tell stars, planets and moons apart | ○ | ○ | ○ |
| name the planets in order | ○ | ○ | ○ |
| explain what keeps planets in orbit | ○ | ○ | ○ |
| make a model solar system | ○ | ○ | ○ |
| name scientists who studied space | ○ | ○ | ○ |